

Yating Wu

Ph.D.

LLM Agents, LLM Reasoning, AI for Science

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Summary

I develop LLM reasoning and agent methods for structuring evidence, tracking information flow, and supporting scientific discovery across language, multimodal data, and tools.

- 🏆 **Selected recognitions:** MIT EECS Rising Stars; EMNLP 2024 Outstanding Paper Award (Top 0.4%).
- 💡 **Pretraining/Post-training:** astronomy-related foundation models, learning signals, and evaluation methods for LLM/VLM reasoning.
- 💡 **LLM Agents:** long-context memory, planning, tool use, and dependency-aware context.
- 💡 **LLM Reasoning:** reasoning structure, information flow, and evidence grounding across text, figures, and multimodal outputs via Questions Under Discussion (QUD).

Education

- 2022 – 2026 **Ph.D. in Electrical and Computer Engineering**, *The University of Texas at Austin*
Advisors: Junyi Jessy Li, Alexandros G. Dimakis
- 2020 – 2022 **M.S. in Electrical and Computer Engineering**, *The University of Texas at Austin*
- 2014 – 2019 **B.Eng. in Computer Science & B.A. in Japanese**, *Dalian University of Technology*
- 2017 - 2018 **Exchange student in Computer Science**, *The University of Tokyo*
Advisor: Toshihiko Yamasaki

Work Experience

- Jun - Oct. 2025 **Applied Scientist Intern**, *AWS AI Labs*, Santa Clara, CA
 - Built long-context memory support for LLM coding agents, including dependency-structured context construction for multi-step tool use
- May - Aug. 2024 **Applied Scientist Intern**, *Amazon Alexa AGI*, Sunnyvale, CA
 - Enhanced planning and tool-use capabilities in LLM agents through question-guided reasoning and agent evaluation

Papers Under Submission and Preprints

- [1] **Yating Wu**, Yuhao Zhang, Sayan Ghosh, Sourya Basu, Anoop Deoras, Jun Huan, Gaurav Gupta “ContextWeaver: Selective and Dependency-Structured Memory Construction for LLM Agents.” Under Submission, 2026. [\[Paper\]](#)

- [2] **Yating Wu**, William Rudman, Venkata S Govindarajan, Alexandros G. Dimakis, Junyi Jessie Li “Multimodal QUD: Inquisitive Questions from Scientific Figures.” Under Submission, 2026. [\[Paper\]](#)

Publications

- [3] Ramya Namuduri, **Yating Wu**, Anshun Asher Zheng, Manya Wadhwa, Greg Durrett and Junyi Jessie Li “QUDsim: Quantifying Discourse Similarities in LLM-Generated Text.” Conference on Language Modeling (**COLM**), 2025 [\[Paper\]](#)
- [4] **Yating Wu***, Ritika Rajesh Mangla*, Alexandros G. Dimakis, Greg Durrett, Junyi Jessie Li. “Which questions should I answer? Saliency Prediction of Inquisitive Questions.” Conference on Empirical Methods in Natural Language Processing (**EMNLP Oral**), 2024. **Outstanding Paper Award (Top 0.4%)**. [\[Paper\]](#)
- [5] Negin Raoof*, **Yating Wu***, Carlos Bonilla*, Junyi Jessie Li, Stephanie M Grasso, Alex Dimakis, Zoi Gkalitsiou. “Modeling Bilingual Disfluencies with Large Language Models.” Workshop on LLMs and Cognition in International Conference on Machine Learning (**ICML Workshop**), 2024. [\[Paper\]](#)
- [6] **Yating Wu**, Ritika Rajesh Mangla, Greg Durrett, Junyi Jessie Li. “QUDeval: The Evaluation of Questions Under Discussion Discourse Parsing.” Conference on Empirical Methods in Natural Language Processing (**EMNLP Oral**), 2023. [\[Paper\]](#)
- [7] **Yating Wu***, William Sheffield*, Kyle Mahowald, and Junyi Jessie Li. “Elaborative Simplification as Implicit Questions Under Discussion.” Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2023. [\[Paper\]](#)
- [8] Wei-Jen Ko, **Yating Wu**, Cutter Dalton, Dananjay Srinivas, Greg Durrett and Junyi Jessie Li. “Discourse Analysis via Questions and Answers: Parsing Dependency Structures of Questions Under Discussion.” Findings of the Association for Computational Linguistics (**ACL**), 2023. [\[Paper\]](#)
- [9] Venelin Kovatchev, Trina Chatterjee, Venkata S Govindarajan, Jifan Chen, Eunsol Choi, Gabriella Chronis, Anubrata Das, Katrin Erk, Matthew Lease, Junyi Jessie Li, **Yating Wu**, Kyle Mahowald. “longhorns at DADC 2022: How many linguists does it take to fool a Question Answering model? A systematic approach to adversarial attacks.” DADC Workshop in The Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL Workshop**), 2022 [\[Paper\]](#)

Technical Skills

AI/ML	Pretraining/post-training, LLM/VLM evaluation, long-context agents, multi-modal AI for Science
Tools	Python, PyTorch, Hugging Face, AWS, CUDA, Git